spikeEncode — decisions needed from Oliver

8 September 2026 · companion to the encoder survey, version 2 · commit 79c503a

Where the work is. All six candidate encoders are implemented and tested, and the probe harness now runs all three tasks end to end with the non-spiking reference measured alongside. That is weeks 1–2 and 5–7 of the twelve-week plan, at day 19.

The catch. Every number so far is on a synthetic stand-in corpus, because there is still no TIMIT. The stand-in has known ground truth, which is what lets the pipeline controls actually fail — and they have, catching two specification errors that would have been invisible on real audio. But it cannot answer what the preliminary experiments exist to ask, and control C1 — the one saying the pipeline is calibrated rather than merely self-consistent — needs a real corpus.

	Decision	What it is holding up
O2	TIMIT licence — does Manchester hold current LDC membership?	The largest blocker in the project: control C1, P1's real answer, the week-4 decision gate, and the weeks 1–2 deliverable. Nineteen days with nothing logged; §9 names it the top risk because it is week-one work.
Q07	Release format — are ON and OFF separate channel indices, or a polarity bit?	The event file format, and controls C6 and C7 with it. Best settled before the featurisation is fixed, since doubling the channel count changes what the decoder receives. Interacts with the SHD convention, which is unipolar.
O3	Spiketrum — the approach to Wijekoon.	Whether E7 enters the comparison at all. Separately: the Spiketrum papers have not been read in full, and the survey's description is from abstracts.
Q28	Do we make the stand-in corpus more speech-like, or wait for TIMIT?	Whether further preliminary work is worth doing. Formant transitions would let P1 and P2 rehearse properly; waiting keeps the battery question on real speech.

Not blocking today: O1 corpus specifics, needed before stage two; O4 whether an IOP read-and-publish agreement covers the article charge; O5 alignment strategy, which depends on O1.

One result worth a minute of the meeting. The study's central engineering question has its first number, and it does not favour the answer the field assumes. The mel reference costs 128,000 bit/s; the spiking encoder reaches 98.5 per cent of its accuracy only at **398,929 bit/s, three times the representation it is approximating**. The two cross near a fifth of that rate, where the encoder sits at ~92 per cent of the bound. Both figures depend on declared bit widths and the corpus is synthetic, so this is not yet a result — but it is the shape of the argument the paper will have to make.

For information: 20 questions are open against the design session, up from nine two days ago. Expected shape rather than drift. An encoder arrives with a specification section and known-answer tests written from its equations, so it either passes or it does not; the harness has neither, the specification declaring the pipeline deliberately unspecified, so it generates questions instead of consuming answers. Six of the twenty are one question, and none needs Oliver.

Nothing above blocks the next fortnight except O2. With a corpus the harness runs as it stands and the first real Pareto fronts follow within days. Without one, what remains is consolidation rather than progress.